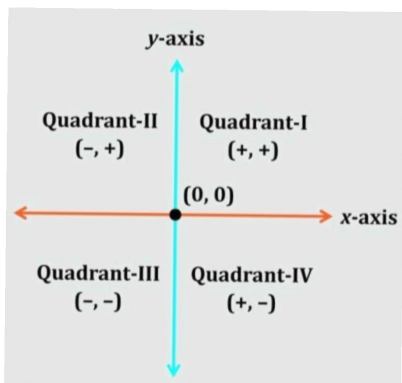
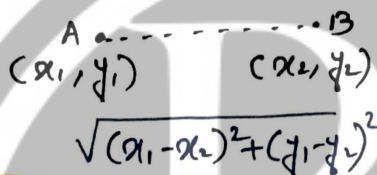


Straight Lines -

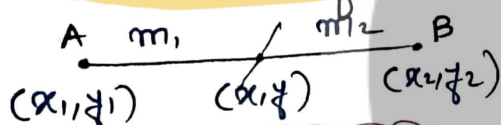


⊕ Distance formula -

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$



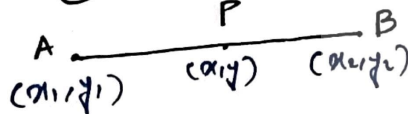
⊕ Section formula -



$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$$

$$y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

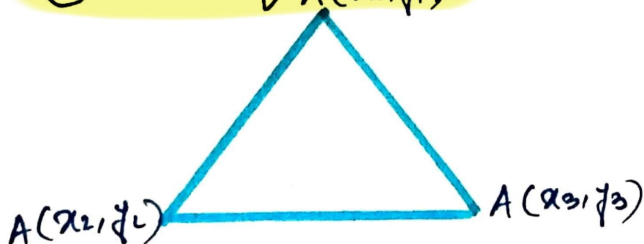
⑧ Mid-point formula -



$$x = \frac{x_1 + x_2}{2}$$

$$\bar{y} = \frac{y_1 + y_2}{2}$$

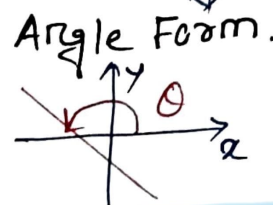
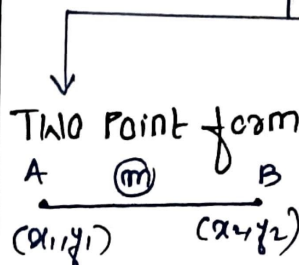
⑧ Area of Triangle -
A(x₁, y₁)



$$\text{Area of } \Delta = \frac{1}{2} \left\{ x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \right\}$$

⑧ Slope of a Line.

TWO ways



Slope of a line $m = \frac{y_2 - y_1}{x_2 - x_1}$

* Slope of line = $m = \tan \theta$

Q. θ = Angle made with the +ve x-axis.

Example) Find the slope of the lines
passing through the points $(3, -2)$ and
 $(-1, 4)$

(-1,4)
b) Passing through the points (3,-2) and (7,-2)
c) Passing through the points (3,-2) and (3,4)

dy Making inclination of 60° with the positive direction of x-axis.

Ans) (i) $m = \frac{y_2 - y_1}{x_2 - x_1} \Rightarrow m = \frac{4 - (-2)}{-1 - 3} = \frac{4+2}{-4}$

$$m = \frac{6}{-4} = -\frac{3}{2} \text{ Ans.}$$

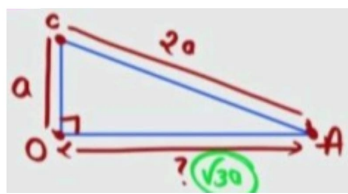
(ii)  $m = \frac{y_2 - y_1}{x_2 - x_1}$
 $= \frac{-2 - (-2)}{7 - 3}$

d)  $\theta = 60^\circ$ $m = \tan \theta$ $m = \tan 60^\circ$ $m = \sqrt{3}$ Ans.

⊗ Point on the x -axis $\Rightarrow (x, 0)$

* Point on the y-axis $\Rightarrow (0, y)$

④ $B = (0, -1)$



Pythagorean Theorem:

$$(H)^2 = (B)^2 + (P)^2$$

$$(AC)^2 = (OC)^2 + (OA)^2$$

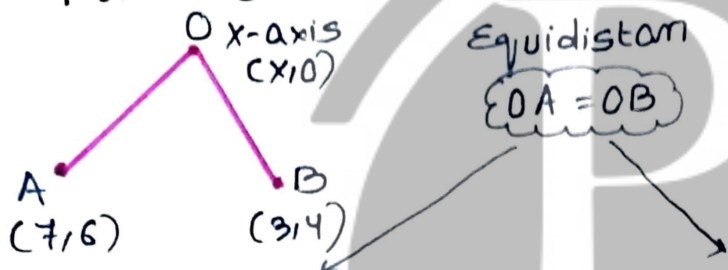
$$(2a)^2 = a^2 + (OA)^2$$

$$4a^2 - a^2 = (OA)^2$$

$$(OA)^2 = 3a^2$$

$$OA = \sqrt{3}a$$

Q. Find a point on the x-axis which is equidistant from the points (7,6) and (3,4)



$$\Rightarrow (\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2})^2 = (\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2})^2$$

$$\Rightarrow (x_1 - x_2)^2 + (y_1 - y_2)^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$

$$\Rightarrow (x - 7)^2 + (0 - 6)^2 = (x - 3)^2 + (0 - 4)^2$$

$$\Rightarrow x^2 + 49 + 14x + 36 = x^2 + 9 - 6x + 16$$

$$\Rightarrow 49 + 36 - 9 + 16 = -6x + 14x$$

$$\Rightarrow 85 - 25 = -6x + 14x$$

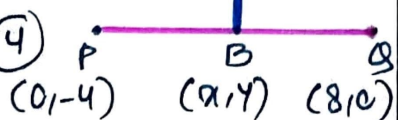
$$\Rightarrow 60 = 8x$$

$$\Rightarrow x = \frac{60}{8} = 7.5$$

Q. Find the slope of a line which passes through the origin and the mid-point of the line segment joining the points P(0,-4) and B(8,0).

$\therefore B$ is a Midpoint (PQ)

$$x = \frac{x_1 + x_2}{2} = \frac{0 + 8}{2} = 4$$



$$y = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-4 - 0}{4 - 0}$$

$$= -1$$

A	slope	B
(x_1, y_1)		(x_2, y_2)
$(0, 0)$		$(4, -2)$

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-2 - 0}{4 - 0}$$

$$= \frac{-2}{4} = -\frac{1}{2}$$

Q. Find the slope of the line which makes an angle 30° with the positive direction of y-axis measured anticlockwise.

Slope of line AB (m_{AB})

$$= \tan \theta$$

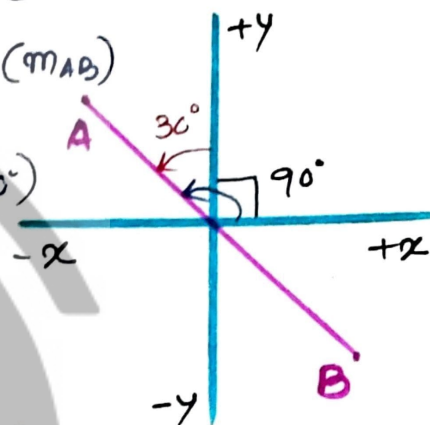
$$\Rightarrow \tan(90 + 30^\circ)$$

$$\Rightarrow \tan 120^\circ$$

$$\Rightarrow \tan(90 + 30^\circ)$$

$$\Rightarrow -\cot 30^\circ$$

$$\Rightarrow -\sqrt{3} \text{ Ans. } \left\{ \begin{array}{l} \tan(180 - 60) \\ -\tan 60 = -\sqrt{3} \end{array} \right.$$



Q. The slope of a line is double of the slope of another line. If tangent of the angle between them is $\frac{1}{3}$, find the slopes of the lines.

$$\tan \theta = \frac{1}{3}$$

$$* \tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

$$\frac{1}{3} = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

$$\pm \frac{1}{3} = \frac{m_2 - m_1}{1 + m_1 m_2}$$

$$\Rightarrow \frac{1}{3} = \frac{m_2 - m_1}{1 + m_1 m_2}$$

$$\Rightarrow \frac{1}{3} = \frac{2m - m}{1 + (m)(2m)}$$

$$\Rightarrow 1 + 2m^2 = 6m - 3m$$

$$\Rightarrow 2m^2 + 1 = 3m$$

$$\Rightarrow 2m^2 - 3m + 1 = 0$$

$$\Rightarrow 2m^2 - 2m - m + 1 = 0$$

$$\Rightarrow 2m(m-1) - 1(m-1) = 0$$

$$\Rightarrow (m-1)(2m-1) = 0$$

$$\Rightarrow m = 1 \text{ or } m = \frac{1}{2}$$

$$\Rightarrow \left[-\frac{1}{3} = \frac{m_2 - m_1}{1 - m_1 m_2} \right]$$

$$\Rightarrow -\frac{1}{3} = \frac{2m - m}{1 + (m)(2m)}$$

$$\Rightarrow -(1 + 2m^2) = 3m$$

$$\Rightarrow -1 - 2m^2 = 3m$$

$$\Rightarrow 0 = 2m^2 + 3m + 1$$

$$\Rightarrow 0 = 2m^2 + 2m + m + 1$$

$$\Rightarrow 0 = 2m(m+1) + 1(m+1)$$

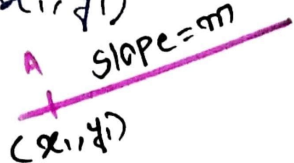
$$m = -1$$

$$m = -1/2$$

Equation of Line: $ax + by + c = 0$

① Slope Point form

Line with slope $= m$,
Passing through Point
(x_1, y_1)

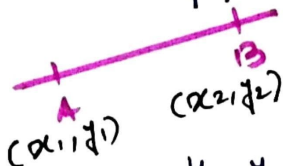


Equation of Line:

$$y - y_1 = m(x - x_1)$$

② Two Point form

Line Passing through
two points (x_1, y_1)
and (x_2, y_2)

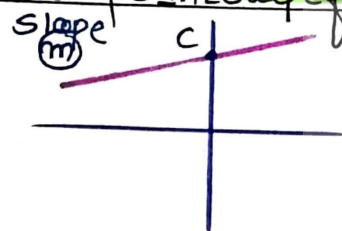


$$\text{slope}(m) = \frac{y_2 - y_1}{x_2 - x_1}$$

Equation of Line:

$$(y - y_1) = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

③ Slope Intercept form ④ Intercept form



Eqn of line:

$$y = mx + c$$

slope Intercept



Eqn of line:

$$\frac{x}{a} + \frac{y}{b} = 1$$

$a = x$ intercept
 $b = y$ intercept.

Q. Eqn. of a line Passing through
the Point $(-4, 3)$ with slope $\frac{1}{2}$.

① Slope point form

$$\text{Eqn of a line: } y - y_1 = m(x - x_1)$$

$$y - 3 = \frac{1}{2}(x + 4)$$

$$2(y - 3) = x + 4$$

$$2y - 6 = x + 4$$

$$2y - x = 10$$

Q. Eqn of a line Passing through
($2, 2\sqrt{3}$) and inclined with the
x-axis at an angle of 75° .

$$y - y_1 = m(x - x_1)$$

$$y - 2\sqrt{3} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} (x - 2)$$

$$(\sqrt{3} - 1)(y - 2\sqrt{3}) = (\sqrt{3} + 1)(x - 2)$$

$$\sqrt{3}y - 6 - y + 2\sqrt{3} = \sqrt{3}x - 2\sqrt{3} + x - 2$$

$$\sqrt{3}y - y - \sqrt{3}x - x = -2 + 6$$

$$\sqrt{3}y - y - \sqrt{3}x - x = 4$$

$$\tan(A + B) = \frac{\tan A \tan B}{1 - \tan A \tan B}$$

$$m = \tan \theta$$

$$m = \tan 75^\circ$$

$$m = \tan(45^\circ + 30^\circ)$$

$$m = \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ}$$

$$m = \frac{1 + \frac{1}{\sqrt{3}}}{1 - (1)(\frac{1}{\sqrt{3}})} \quad m = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$$

$$m = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$$

Q. The Vertices of ΔPQR are $P(2, 1)$,
 $Q(-2, 3)$ and $R(4, 5)$. Find equation
of the median through the vertex R .

OR is Median:
Coordinate of $O \rightarrow$ Median of PQ

$$x = \frac{x_1 + x_2}{2} \quad y = \frac{y_1 + y_2}{2}$$

$$x = \frac{-2 + 2}{2} \quad y = \frac{3 + 1}{2}$$

$$x = 0 \quad y = 2$$

$$O(0, 2) \quad R(4, 5)$$

$$x_1, y_1 \quad x_2, y_2$$

$$\text{Eqn of a line: } y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$y - 2 = \frac{5 - 2}{4 - 0} (x - 0)$$

$$y - 2 = \frac{3}{4}x$$

$$4y - 8 = 3x$$

$$4y - 3x = 8$$

Ques) Find equation of the line passing through the point (2,2) and cutting off intercepts on the axes whose sum is 9.

Given: $a+b=9 \rightarrow b=9-a$

$x=2, y=2$

$\Rightarrow \frac{x}{a} + \frac{y}{b} = 1$

$\Rightarrow \frac{2}{a} + \frac{2}{9-a} = 1$

$\Rightarrow \frac{2(9-a) + 2a}{a(9-a)} = 1$

$\Rightarrow 18 - 2a + 2a = 1(90 - a^2) \therefore b=9-a$

$\Rightarrow 18 = 9a - a^2$

$\Rightarrow 18 = 9a - a^2$

$\Rightarrow a^2 - 9a + 18 = 0$

$\Rightarrow a^2 - 6a - 3a + 18 = 0$

$\Rightarrow a(a-6) - 3(a-6) = 0$

$\Rightarrow (a-6)(a-3) = 0$

$a=6 \quad a=3$

$a=6 \quad b=9-a \quad (b=3)$

$a=3 \quad b=9-3 \quad (b=6)$

Now Eqn of Line: $\frac{x}{a} + \frac{y}{b} = 1$

$a=6, b=3$

$\frac{x}{6} + \frac{y}{3} = 1$

$a=3, b=6$

$\frac{x}{3} + \frac{y}{6} = 1$

Various forms of Eqn of Line -

$Ax + By + C = 0$

Slope Intercept form

Intercept form

$Ax + By + C = 0$

$\hookrightarrow By = -C - Ax$

$\hookrightarrow y = \frac{-Ax - C}{B}$

$\hookrightarrow y = \frac{-A}{B}x - \frac{C}{B}$

Slope = $-\frac{A}{B}$

y intercept = $-\frac{C}{B}$

$Ax + By + C = 0$

$Ax + By = -C$

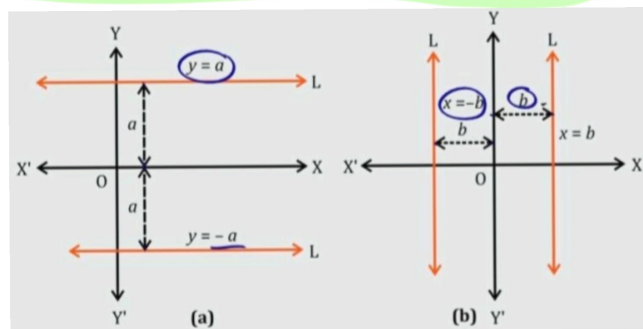
$\frac{Ax + By}{-C} = 1$

$\frac{Ax}{-C} + \frac{By}{-C} = 1$

$\frac{x}{-C/A} + \frac{y}{-C/B} = 1$

$\left\{ \begin{array}{l} x \text{ intercept} = -C/A \\ y \text{ intercept} = -C/B \end{array} \right.$

Horizontal & Vertical Lines



2. Point-Slope Form:

Line With slope = m, Passing through Point (x_1, y_1) .

$y - y_1 = m(x - x_1)$

Distance of a Point from a line on the x-axis.

Find the Points whose dist from the line $\frac{x}{3} + \frac{y}{4} = 1$ are 4 units

Line = $4x + 3y = 12$

$\hookrightarrow 4x + 3y - 12 = 0$

Point = $(x, 0)$

Distance = 4 units.

$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$

$4 = \frac{|(4)(x) + (3)(0) + (-12)|}{\sqrt{4^2 + 3^2}}$

$4 = \frac{|4x - 12|}{\sqrt{16 + 9}}$

$4 = \frac{|4x - 12|}{5}$

$\Rightarrow 20 = |4x - 12|$

$\Rightarrow \pm 20 = 4x - 12$

$+20 = 4x - 12$

$20 + 12 = 4x$

$32 = 4x$

$x = 8$

$-20 = 4x - 12$

$-20 + 12 = 4x$

$-8 = 4x$

$x = -2$

$(8, 0)$ or $(-2, 0)$
Final Answer

Distance of a Point from a line -

- Line: $Ax + By + C = 0$
- Point: (x_1, y_1)

To find Perpendicular distance (d) from point to line, use the formula:

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

- Absolute value in the numerator makes sure distance is positive.
- Denominator is the length of the vector perpendicular to the line.

Quest) Reduce the following equations into slope-intercept form and find their slopes and the y-intercept.

i) $x + 7y = 0$ ii) $3x + 2y - 12 = 0$ iii) $y = 0$

$x + 7y = 0$
 $Ax + By + C = 0$

$A = 1, B = 7$
 slope = $-A/B$
 $= -1/7$

* $\frac{x}{a} + \frac{y}{b} = 1$

$x + 7y = 1$

$\frac{x}{1} + \frac{y}{1/7} = 1$

x-intercept y-intercept.

Distance between Two Parallel Lines -

Let the lines be:

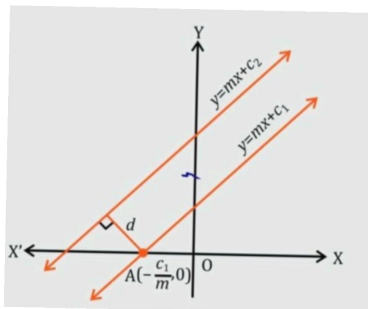
Let the lines be:

$y = mx + C_1$

$y = mx + C_2$

Then, dist b/w them:

$$d = \frac{|C_1 - C_2|}{\sqrt{1 + m^2}}$$



If lines are in General form:

$Ax + By + C_1 = 0$

$Ax + By + C_2 = 0$

Then distance:

$$d = \frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}$$

Quest) Find the distance b/w

Parallel Lines:

i) $(15x + 8y) - 34 = 0$ and $(15x + 8y) + 31 = 0$

$A_1x + B_1y + C_1 = 0$

$A_2x + B_2y + C_2 = 0$

$A_1 = 15, B_1 = 8, C_1 = -34, A_2 = 15, B_2 = 8, C_2 = 31$

$d = \frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}} = \frac{|-34 - 31|}{\sqrt{(15)^2 + (8)^2}} = \frac{|-65|}{\sqrt{225 + 64}}$

$d = \frac{65}{\sqrt{289}} = \frac{65}{17}$ Ans.

ii) $L(x+y) + P = 0$ and $L(x+y) - r = 0$

$1x + 1y + P = 0$

$1x + 1y - r = 0$

$C_1 = P$

$C_2 = -r$

$d = \frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}} = \frac{|P - (-r)|}{\sqrt{1^2 + 1^2}} = \frac{P + r}{\sqrt{2}}$

$d = \frac{1}{\sqrt{2}} \left(\frac{P + r}{1} \right)$ Ans.

Quest) Find Angles between the lines

$\sqrt{3}x + y = 1$ and $x + \sqrt{3}y = 1$

slope = $-A/B$

slope = $-A/B = -1/\sqrt{3}$

$m_1 = -\sqrt{3}$

$m_2 = -1/\sqrt{3}$

$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| = \left| \frac{-1/\sqrt{3} - (-\sqrt{3})}{1 + (-\sqrt{3})(-1/\sqrt{3})} \right|$

$= \left| \frac{-1 + (\sqrt{3})(\sqrt{3})}{1 + 1} \right| = \left| \frac{-1 + 3}{2} \right|$

$= \frac{2/\sqrt{3}}{2} = \frac{2}{\sqrt{3}} \times \frac{1}{2} = \frac{1}{\sqrt{3}}$

$\theta = 30^\circ$ Ans.

Quest) The line through the points $(h, 3)$ and $(4, 1)$ intersects the line $7x - 9y - 19 = 0$ at right angle. Find the value of h.

i) $(h, 3)$

$(4, 1)$

ii) $7x - 9y - 19 = 0$

$$\text{Slope}(m_1) = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1-3}{4-h} \Rightarrow m_1 = \frac{-2}{4-h}$$

$$\text{Slope}(m_2) = -A/B = -7/-9 \Rightarrow m_2 = 7/9$$

Right Angle $\Rightarrow$ 2 Lines $\perp$ ($J_1 \perp J_2$)

$$\therefore m_1 \times m_2 = -1$$

$$\Rightarrow \frac{-2}{(4-h)} \times \frac{7}{9} = -1$$

$$\Rightarrow \frac{14}{9(4-h)} = 1$$

$$\Rightarrow 14 = 36 - 9h$$

$$\Rightarrow 9h = 36 - 14 = 22$$

$$\Rightarrow h = \frac{22}{9} \text{ Ans.}$$

Quest) Find the co-ordinates of the foot of Perpendicular from the Point $(-1, 3)$ to the line $3x - 4y - 16 = 0$

Eqn of line (J_1):

$$y - y_1 = m_1(x - x_1)$$

$$y - 3 = \frac{-4}{3}(x + 1)$$

$$3y - 9 = -4(x + 1)$$

$$3y - 9 = -4x - 4$$

$$4x + 3y = -4 + 9$$

$$4x + 3y = 5 \quad \text{--- (i)}$$

$J_1 \perp J_2$

$$\therefore m_1 \times m_2 = -1$$

$$m_1 \times \frac{3}{4} = -1$$

$$m_1 = -4/3$$

$$3(4x + 3y = 5)$$

$$4(3x - 4y = 16)$$

$$\Rightarrow 12x + 9y = 15$$

$$\begin{array}{r} 12x + 9y = 15 \\ (-) 12x - 16y = 64 \\ \hline (-) 25y = -49 \end{array}$$

$x = ? \quad y = ?$ (calculate).

Quest) The Perpendicular from the origin to the line $y = mx + c$ meets it at the Points $(-1, 2)$. Find the values of m and c .

$$(J_1) \quad (0, c)$$

$$(J_2)$$

$$(-1, 2)$$

$$y = mx + c$$

$$y = mx + c$$

$$2 = 2(-1) + c$$

$$2 = -2 + c$$

$$c = 4$$

$$\text{Line } (J_1) \Rightarrow \text{slope}(m_1) = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2-0}{-1-0}$$

$$m_1 = -\frac{1}{2}$$

$$J_1 \perp J_2$$

$$m_1 \times m_2 = -1$$

$$-\frac{1}{2} \times m_2 = -1$$

$$m_2 = 2$$

$$\begin{cases} x = -1 \\ y = 2 \\ m_2 = 2 \\ c = 4 \end{cases}$$

Quest) If P is length of Perpendicular from the origin to the line whose intercepts on the axes are a & b then show that

$$\frac{1}{P^2} = \frac{1}{a^2} + \frac{1}{b^2}$$

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

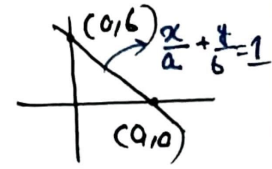
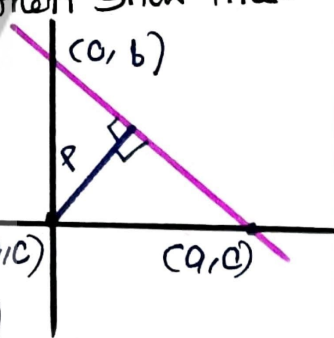
$$P = \frac{|A(0) + B(0) + (-1)|}{\sqrt{(\frac{1}{a})^2 + (\frac{1}{b})^2}}$$

$$P = \frac{|-1|}{\sqrt{(\frac{1}{a})^2 + (\frac{1}{b})^2}}$$

$$P = \frac{1}{\sqrt{1/a^2 + 1/b^2}}$$

sq. both sides

$$P^2 = \frac{1^2}{1/a^2 + 1/b^2} \Rightarrow \frac{1}{P^2} = \frac{1}{a^2} + \frac{1}{b^2}$$



$$A = \frac{1}{a} \quad B = \frac{1}{b}$$